 

# NAGASAKI, Japan

WMO: **478170**

Lat: **32.7343N** Lon: **129.8680E** Elev: **36** StdP: **100.89** Time Zone: **9.00 (E09)** Period: **94-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **1** | **0.8** | **1.8** | **-7.3** | **2.0** | **4.9** | **-6.0** | **2.3** | **5.0** | **8.8** | **8.5** | **7.8** | **9.6** | **2.2** | **80** | **0.355** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **8** | **5.7** | **33.0** | **25.8** | **32.0** | **25.7** | **31.1** | **25.4** | **27.1** | **30.5** | **26.6** | **29.9** | **26.2** | **29.4** | **3.2** | **230** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **26.2** | **21.6** | **28.8** | **25.7** | **21.0** | **28.5** | **25.3** | **20.5** | **28.4** | **85.5** | **30.7** | **83.4** | **30.0** | **81.6** | **29.4** | **28.8** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **7.6** | **6.4** | **5.5** | DB | **-1.2** | **35.4** | **1.0** | **1.0** | **-1.9** | **36.1** | **-2.5** | **36.7** | **-3.1** | **37.3** | **-3.8** | **38.0** | (4) |
| (5) | | | | WB | **-2.8** | **27.4** | **1.0** | **0.7** | **-3.5** | **27.9** | **-4.1** | **28.3** | **-4.6** | **28.7** | **-5.3** | **29.2** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | DBAvg | | **17.7** | **7.1** | **8.2** | **11.2** | **15.7** | **19.9** | **23.2** | **27.4** | **28.6** | **25.4** | **20.4** | **14.7** | **9.4** | (6) |
| (7) | | DBStd | | **7.73** | **2.77** | **3.15** | **3.03** | **2.74** | **1.95** | **1.89** | **1.97** | **1.68** | **2.27** | **2.61** | **3.18** | **3.04** | (7) |
| (8) | | HDD10.0 | | **236** | **97** | **68** | **21** | **1** | **0** | **0** | **0** | **0** | **0** | **0** | **2** | **48** | (8) |
| (9) | | HDD18.3 | | **1344** | **348** | **284** | **220** | **85** | **7** | **0** | **0** | **0** | **0** | **10** | **115** | **276** | (9) |
| (10) | | CDD10.0 | | **3031** | **7** | **17** | **59** | **172** | **308** | **397** | **538** | **575** | **461** | **323** | **143** | **31** | (10) |
| (11) | | CDD18.3 | | **1096** | **0** | **0** | **0** | **7** | **56** | **147** | **279** | **317** | **211** | **74** | **6** | **0** | (11) |
| (12) | | CDH23.3 | | **9264** | **0** | **0** | **0** | **11** | **137** | **673** | **2862** | **3677** | **1639** | **262** | **3** | **0** | (12) |
| (13) | | CDH26.7 | | **2891** | **0** | **0** | **0** | **1** | **7** | **78** | **921** | **1448** | **411** | **26** | **0** | **0** | (13) |
| (14) | Wind | WSAvg | | **2.3** | **2.3** | **2.4** | **2.6** | **2.5** | **2.2** | **2.2** | **2.5** | **2.2** | **2.1** | **2.0** | **2.0** | **2.2** | (14) |
| (15) | Precipitation | PrecAvg | | **1920** | **70** | **84** | **119** | **167** | **180** | **331** | **300** | **199** | **189** | **103** | **93** | **70** | (15) |
| (16) | | PrecMax | | **2846** | **237** | **192** | **350** | **348** | **390** | **789** | **1179** | **668** | **490** | **280** | **214** | **152** | (16) |
| (17) | | PrecMin | | **924** | **9** | **22** | **43** | **36** | **50** | **44** | **6** | **7** | **2** | **4** | **12** | **5** | (17) |
| (18) | | PrecStd | | **390** | **38** | **42** | **51** | **66** | **80** | **157** | **217** | **136** | **110** | **70** | **52** | **34** | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | **17.0** | **19.1** | **21.2** | **24.7** | **27.6** | **29.9** | **33.9** | **35.0** | **32.5** | **29.2** | **23.9** | **18.8** | (19) |
| (20) | | | MCWB | | **14.2** | **15.4** | **15.5** | **17.9** | **20.0** | **23.6** | **25.6** | **25.8** | **24.8** | **22.4** | **18.9** | **15.4** | (20) |
| (21) | | 2% | DB | | **14.9** | **17.2** | **19.3** | **22.8** | **25.9** | **28.5** | **32.4** | **33.5** | **31.2** | **27.2** | **22.2** | **17.2** | (21) |
| (22) | | | MCWB | | **11.9** | **14.2** | **14.6** | **16.9** | **19.1** | **23.2** | **26.0** | **25.7** | **24.6** | **21.2** | **17.6** | **13.3** | (22) |
| (23) | | 5% | DB | | **13.2** | **15.3** | **17.9** | **21.7** | **24.8** | **27.5** | **31.4** | **32.6** | **30.1** | **25.9** | **20.9** | **15.8** | (23) |
| (24) | | | MCWB | | **9.6** | **12.1** | **13.8** | **16.4** | **18.8** | **23.0** | **25.7** | **25.8** | **23.9** | **20.1** | **16.9** | **12.0** | (24) |
| (25) | | 10% | DB | | **11.7** | **13.5** | **16.5** | **20.4** | **23.8** | **26.6** | **30.5** | **31.7** | **29.0** | **24.7** | **19.7** | **14.4** | (25) |
| (26) | | | MCWB | | **8.1** | **9.8** | **12.7** | **15.7** | **18.4** | **22.6** | **25.5** | **25.7** | **23.5** | **19.3** | **15.8** | **11.0** | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | **15.4** | **16.7** | **17.6** | **20.9** | **23.1** | **25.9** | **27.8** | **27.7** | **26.7** | **24.7** | **20.6** | **17.1** | (27) |
| (28) | | | MCDB | | **16.6** | **17.9** | **19.2** | **22.7** | **25.0** | **28.1** | **31.0** | **31.2** | **29.6** | **26.9** | **22.2** | **18.1** | (28) |
| (29) | | 2% | WB | | **12.6** | **14.8** | **16.0** | **19.1** | **21.7** | **25.2** | **27.0** | **27.2** | **25.9** | **23.2** | **19.2** | **14.3** | (29) |
| (30) | | | MCDB | | **14.3** | **16.7** | **18.1** | **21.1** | **23.5** | **27.3** | **30.2** | **30.8** | **28.9** | **25.8** | **20.9** | **16.2** | (30) |
| (31) | | 5% | WB | | **10.2** | **12.6** | **14.6** | **17.9** | **20.6** | **24.6** | **26.5** | **26.7** | **25.3** | **21.9** | **17.8** | **12.8** | (31) |
| (32) | | | MCDB | | **12.4** | **14.9** | **17.2** | **20.2** | **22.8** | **26.6** | **29.5** | **30.3** | **28.5** | **24.6** | **20.2** | **15.0** | (32) |
| (33) | | 10% | WB | | **8.7** | **10.4** | **13.2** | **16.7** | **19.6** | **23.7** | **26.1** | **26.4** | **24.6** | **20.4** | **16.5** | **11.4** | (33) |
| (34) | | | MCDB | | **11.2** | **12.8** | **16.1** | **19.4** | **22.5** | **25.6** | **29.0** | **30.0** | **27.9** | **23.5** | **19.1** | **14.0** | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | **6.1** | **6.8** | **7.1** | **7.3** | **7.1** | **5.5** | **5.1** | **5.7** | **6.2** | **6.9** | **6.8** | **6.4** | (35) |
| (36) | | 5% DB | MCDBR | | **8.1** | **8.7** | **8.8** | **8.8** | **8.4** | **6.7** | **6.3** | **6.8** | **6.9** | **7.8** | **7.5** | **7.9** | (36) |
| (37) | | | MCWBR | | **6.1** | **6.7** | **6.1** | **5.0** | **4.3** | **3.2** | **2.4** | **2.4** | **2.9** | **3.7** | **4.6** | **5.5** | (37) |
| (38) | | 5% WB | MCDBR | | **7.6** | **7.7** | **7.4** | **6.6** | **6.0** | **4.5** | **5.4** | **5.8** | **5.8** | **6.0** | **6.2** | **7.0** | (38) |
| (39) | | | MCWBR | | **6.7** | **7.3** | **6.6** | **5.2** | **4.4** | **2.8** | **2.4** | **2.4** | **2.8** | **3.8** | **4.9** | **6.0** | (39) |
| (40) | Clear-Sky Solar Irradiance | taub | | | **0.445** | **0.484** | **0.543** | **0.562** | **0.571** | **0.582** | **0.564** | **0.539** | **0.512** | **0.468** | **0.438** | **0.434** | (40) |
| (41) | | taud | | | **2.025** | **1.959** | **1.822** | **1.830** | **1.870** | **1.932** | **2.019** | **2.091** | **2.113** | **2.135** | **2.133** | **2.064** | (41) |
| (42) | | Ebn at Noon | | | **737** | **752** | **740** | **749** | **746** | **735** | **746** | **759** | **761** | **762** | **743** | **722** | (42) |
| (43) | | Edh at Noon | | | **139** | **164** | **203** | **210** | **203** | **190** | **174** | **160** | **150** | **136** | **124** | **128** | (43) |
| (44) | All-Sky Solar Radiation | RadAvg | | | **2.17** | **2.93** | **4.03** | **4.84** | **5.15** | **4.33** | **5.17** | **5.38** | **4.36** | **3.63** | **2.61** | **2.02** | (44) |
| (45) | | RadStd | | | **0.22** | **0.29** | **0.27** | **0.42** | **0.59** | **0.67** | **0.71** | **0.47** | **0.45** | **0.36** | **0.26** | **0.16** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (30 neighbors) | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (47) |

Nomenclature: See separate page